GG-S-00783c (DSA-DM)
March 13, 1970
Interim Revision of
Fed. Spec. GG-D-423a
January 3, 1957, and
Fed. Spec. GG-S-783b
February 10, 1958

INTERIM FEDERAL SPECIFICATION

STRIP ASSORTMENT AND DISKS, ALUMINUM OXIDE ABRASIVE, SAFE-SIDED, METAL-BACKED, DENTAL

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specifications GG-D-423a and GG-S-783b.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a safe-sided, aluminum oxide abrasive strip assortment and disks suitable for use during dental polishing procedures.

1.2 Classification. The abrasive strip assortment and disks shall be of the following types and sizes, as specified (see 6.1):

Type I. Strip assortment.
Type II. Disk.

Size 1. 5/8-inch diameter.
Size 2. 7/8-inch diameter.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

STANDARDS

Federal

FED-STD-123	Marking For Domestic Shipment (Civilian Agencies).
FED-STD-151	Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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SPECIFICATION

Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

U.S. DEPARTMENT OF COMMERCE
NATIONAL BUREAU OF STANDARDS
COMMERCIAL STANDARD (CS)

CS 271-65 Grading of Abrasive Grain for Grinding Wheels

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation E 290 Semi-Guided Bend Test for Ductility of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Backing. The backing shall be fabricated from cold-rolled, hard-tempered, low carbon steel with a carbon content of 0.08 to 0.23 percent. After electroplating, the steel shall be stress relieved to remove any hydrogen embrittlement.

3.1.2 Abrasive. The abrasive shall be fabricated from crystalline, aluminum oxide particles which have been electrically furnaced and fused. The particles shall be 180 grit in accordance with the grading requirements of CS 271-65. The abrasive shall be clean and free from dirt, lint, or other foreign materials.

3.1.3 Bonding agent. The bonding agent shall be nickel electroplating.

3.2 Style, design, and dimensions.

3.2.1 Type I. The strips in the strip assortment shall be 5.2 to 5.4 inches long, 0.0020 to 0.0030 inch thick without abrasive, and 0.0060 to 0.0070 inch thick with abrasive. The narrow, medium, and wide strips shall have widths of 0.09 to 0.10 inch, 0.12 to 0.13 inch, and 0.15 to 0.16 inch, respectively. The strips shall be suitable for use during dental polishing procedures.

3.2.2 Type II. The unplated, uncoated metal disk (backing) shall be 0.0040 $\pm$ 0.0005 inch thick, and the plated, coated disk shall be 0.0075 to 0.0100 inch thick. The size 1 disk shall have a diameter of 0.625 $\pm$ 0.012 inch and the size 2 disk shall have a diameter of 0.875 $\pm$ 0.015 inch. The disks shall contain a mandrel hole 0.076 $\pm$ 0.003 inch in diameter, centered in the disk. The disks shall be suitable for attachment to standard dental screw mandrels used with dental handpieces during polishing procedures.

3.3 Construction.

The strip assortment and the disk shall essentially consist of aluminum oxide abrasive particles bonded to one side of a metal backing by nickel electroplating. The backing shall be a flat, smooth, plain strip (Type I) or disk (Type II). Both the safeside surface and the coatside surface of the backing shall be nickel plated. In addition, the coatside surface shall contain a closed coat of the abrasive particles, uniformly distributed and firmly attached to the metal strip or disk by means of additional electroplated nickel.

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3.4 Width assortment. Each strip assortment (Type I) shall consist of 12 strips, 4 of each width (narrow, medium, and wide), as specified in 3.2.1.

3.5 Deterioration. The aluminum oxide abrasive particles shall not flake off, rub off, or chip off, and the metal strip or disk shall show no evidence of deterioration, when the abrasive strips or disk are subjected to one complete cycle of the test specified in 4.4.2.

3.6 Hand bend. Type I strips shall withstand a 180-degree hand bend around a 1/16 inch round pin, when tested as specified in 4.4.4.

3.7 Workmanship. Workmanship shall be first class throughout. The abrasive strips and disks shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. The unit of product for examination purposes shall be one of each size of strip in the assortment (narrow, medium, and wide) for Type I, and one disk of the applicable size for Type II.

Table I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for the deterioration test shall be one of each width of strip for Type I and one disk of the applicable size for Type II. The unit of product for the hand bend test shall be one of each width of strip for Type I.

Table II. Test criteria

Component	Characteristic	Requirement	Test procedure	Inspection level
Raw abrasive	Grading of grit	3.1.2	4.4.1	(see 4.2.2.1)
Finished item	Deterioration	3.5	4.4.2	S-3
Raw steel	Chemical analysis	3.1.1	4.4.3	(see 4.2.2.2)
Strip	Hand bend	3.6	4.4.4	S-3

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4.2.2.1 Sampling for abrasive particle size test. Sampling for the determination of abrasive particle grit size shall be conducted in accordance with the procedures set forth in CS 271-65. Certification from suppliers of the abrasive will be acceptable in lieu of testing.

4.2.2.2 Sampling for chemical analysis. A sufficient amount from each lot of raw metal used for the backing shall be selected for chemical analysis.

4.3 Examination. The strips and disks shall be examined for defects including, but not limited to, those listed in Table III.

Table III. Classification of defects

Category	Defect
Major A:	
101	Style and design not as specified.
102	Abrasive particles do not form a closed coat, or are not uniformly distributed over entire coatside surface.
103	Coatside surface of strip or disk contains dirt, lint, or other foreign matter.
104	Both sides of metal backing coated, partially coated, or not coated at all.
105	Mandrel hole not present in disk, or not properly centered in disk.
Major B:	
151	Overall thickness of abrasive strip or disk not within range specified.
152	Diameter of mandrel hole of disk not within limits specified.
153	Fewer than 4 of each width of strip furnished in strip assortment; fewer than 12 strips per assortment.
Minor A:	
201	Safeside surface of strip or disk not flat and not smooth.
202	Safeside surface of strip or disk not free from dirt, lint, or other foreign material.
203	Packaging and packing not as specified.
204	Marking on packaging and pack not as specified.

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Table III. Classification of Defects (Con't)

Minor B:	
251	Length of strips not as specified.
252	Applicable width size of strip not as specified.
253	Diameter of mandrel hole of disk not as specified.

4.4 Test procedures.

4.4.1 Grading of grit. The grading of the abrasive grit shall be determined in accordance with the test procedures set forth in CS 271-65.

4.4.2 Deterioration.

4.4.2.1 Resistance to sulfuric acid. The abrasive strips or disks shall be completely immersed in a 10 percent aqueous solution of sulfuric acid at 25.0 ± 1.0 degrees Centigrade. At the end of one minute (± 3 seconds), the strips or disks shall be removed from the sulfuric acid solution and thoroughly rinsed in distilled water. Examination shall be made for evidence of deterioration.

4.4.2.2 Resistance to ethyl alcohol. The strips or disks, as applicable, shall then be completely immersed in 190-proof (95 percent by volume) ethyl alcohol at 25.0 ± 1.0 degrees Centigrade. At the end of 4 ± 1/4 hours, the strips or disks shall be removed from the alcohol and examined for evidence of deterioration.

4.4.2.3 Resistance to thermal shock. Next, the strips or disks shall be passed through the flame of an alcohol lamp (such as is used in annealing gold foil), rinsed in distilled water, and then examined a third time for possible deterioration, as specified in 3.5.

4.4.3 Chemical analysis. In the absence of certificates of quality for the raw steel, the material used to fabricate the backing shall be subjected to chemical analysis in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.4 Hand bend test. The hand bend test on the strip assortment shall be performed in accordance with ASTM Designation E 290.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

5.1.1 Unit of issue. One box (12 disks or 12 strips), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Type	Unit package	Intermediate package	Exterior container
I	1 unit	24 units	288 units
II	1 unit	12 units	576 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Strip assortment. Each unit shall be packaged in a paperboard box of appropriate size and design. Closure shall be adequate to prevent accidental opening.

5.1.3.1.2 Disk. Disks shall be wrapped in neutral tissue or other suitable anti-corrosive material in such a manner as to prevent the individual disks from coming in contact with each other. Disks shall be further packaged in a paperboard box of appropriate size and design. Closure shall be adequate to prevent accidental opening.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design, constructed in accordance with PPP-B-566, or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodities indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document-(s) or serve to amend contractual requirements .

SPECIFICATION

GG-S-00783c(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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